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Final
EXAMINATION
December 2017

DURATION: 3 HOURS

No. of Students:

Department Name & Course Number: Mathematics & Statistics MATH2004A,B,C,D,F
Course Instructor(s) X Hua (A, D), M. J. Moore (B), M. Lemire (C), J. Abdulrahman (F).

AUTHORIZED MEMORANDA

NON-PROGRAMMABLE, NON-GRAPHING CALCULATORS PERMITTED

Students MUST count the number of pages in this examination question paper before beginning to write, and report any discrepancy to a proctor. This question paper has 6 pages.

This examination question paper MAY NOT be taken from the examination room.

In addition to this question paper, students require: an examination booklet no
a Scantron sheet Yes
blank paper for rough work Yes

COURSE INSTRUCTOR AND SECTION (See above)

LAST NAME GIVEN NAME

STUDENT # SIGNATURE

ANSWER ALL QUESTIONS

In part A use a pencil to indicate your answer on the Scantron sheet.

In part B you must justify all of your assertions
and show all of your work.

Only Scantron sheets and work on this booklet will be marked but
you must submit your scratch paper also.
If you need additional space to write your answers, continue on the last
page of this paper, noting “continued on page 6”.

Questions	Maximum	Score
A	48	
B1	10	
B2	8	
B3	9	
B4	9	
B5	6	
B6	10	
Total	100	

Multiple choice questions: use a pencil to select your answer on the Scantron sheet. **Only select one answer!**

A1. Let θ be the angle between the two planes $x + 2y - 2z - 3 = 0$ and $2x + 2y - z - 5 = 0$. Find $\cos \theta$.

- (a) $\frac{4}{9}$ (b) $\frac{9}{4}$ (c) $\frac{8}{9}$ (d) $\frac{9}{8}$ (e) $-\frac{4}{9}$

A2. Find the length of the polar curve $r = \cos \theta + \sin \theta$, $0 \leq \theta \leq \pi/4$.

- (a) $\sqrt{2}\pi/4$ (b) $\sqrt{2}\pi/2$ (c) $3\sqrt{2}\pi/4$ (d) $5\sqrt{2}\pi/4$ (e) $\pi/2$.

A3. Find the directional derivative of $f(x, y, z) = z^3 + e^{xyz}$ at the point $(1, 1, 2)$ in the direction of the vector $\underline{v} = (1, 2, 2)$.

- (a) $2e/3$ (b) $5e/3$ (c) 4 (d) 5 (e) 9

A4. Find the tangent plane to the surface $x^3y - y^3x + z^3x = -5$ at the point $P(1, 2, 1)$.

(a) $x+2y+z=6$ (b) $-7x+11y+5z=-20$ (c) $7x-11y-5z=-20$ (d) $7x-13y-11z=-30$ (e) $7x+13y+11z=44$

A5. The volume under the surface $z = 8 - 2x^2 - y^2$ which projects onto the region $\Re = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 2\}$ is

- (a) 8 (b) 5 (c) 12 (d) 4 (e) 1

A6. If the vectors $\underline{v} = 2\underline{i} - \underline{j} + \underline{k}$ and $\underline{w} = 3\underline{i} + 2\underline{j} + 4\underline{k}$ are orthogonal, then $t =$

- (a) 8 (b) -1 (c) 0 (d) 5 (e) 13

A7. Let $w = xyz + x^2 + y^3 + z^4$ and $x = 2s - 3t$, $y = st^2$, $z = s^2 + t^3 - 1$, then $\frac{\partial w}{\partial t} \Big|_{s=1, t=1} =$

- (a) 16 (b) 30 (c) 24 (d) 3 (e) 18

A8. Find $\text{curl } \mathbf{F}$ for the vector field $\mathbf{F}(x, y, z) = (3x^2, 2z, -x) = 3x^2\underline{i} + 2z\underline{j} - x\underline{k}$:

- (a) $5\underline{i} - 4\underline{j} + \underline{k}$ (b) $3\underline{i} - 2\underline{j} + \underline{k}$ (c) $-2\underline{i} + \underline{j}$ (d) $6\underline{k}$ (e) none of these

A9. Given the point $(\sqrt{2}, 0, 1)$ in Cartesian coordinates, its cylindrical coordinates are

- (a) $(\sqrt{2}, 0, 1)$ (b) $(\sqrt{2}, \pi, 1)$ (c) $(\sqrt{3}, 0, 1)$ (d) $(\sqrt{2}, \pi/2, 1)$ (e) none of these

A10. Given the point $(-1, 0, 1)$ in Cartesian coordinates, its spherical coordinates are

- (a) $(1, 0, \pi/4)$ (b) $(\sqrt{2}, 0, \pi/4)$ (c) $(\sqrt{2}, \pi/4, 0)$ (d) $(\sqrt{2}, \pi, \pi/4)$ (e) none of these

A11. The function $f(x, y) = x^2y - x^2 - 2y^2$ has critical points $(0, 0)$ and $(2, 1)$. These are, respectively:

- (a) both maxima (b) both minima (c) a maximum and a minimum
(d) a minimum and a saddle point (e) a maximum and a saddle point

A12. Let E be the region in $\mathbb{R}^3$ above the square $[0, 1] \times [0, 1]$ and below the plane $x+y+z = 6$

$$\iiint_E 6xy dV =$$

- (a) 3 (b) 5 (c) 2 (d) 4 (e) 1

B1. Use Lagrange Multipliers to find the maximum and minimum values of the function $f(x,y) = x^3 - 8y^3$ on the curve $x^2 + 4y^2 = 16$. [10 marks]

B2. Draw the region of integration and evaluate $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$. [8 marks]

B3. Evaluate the double integral $\iint_R (x + 2y) dA$,

where R is the region between $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$ above the x -axis. [9 marks]

B4. Let R be the hemi-spherical shell which is between the two spheres $x^2 + y^2 + z^2 = 36$ and $x^2 + y^2 + z^2 = 49$ and where $z \geq 0$. If the density at (x, y, z) in R is given by $\delta(x, y, z) = 4(x^2 + y^2 + z^2)^{-1/2}$, calculate the mass of R .

[9 marks]

B5. Consider the vector field $F(x,y,z) = (y-x, z-y, y-x) = (y-x)\underline{i} + (z-y)\underline{j} + (y-x)\underline{k}$. Let E be the solid cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$ and let S denote the surface which is the boundary of E , oriented outwards. Use Gauss' Divergence Theorem to compute the surface integral of the vector field F over S : $\iint_S F \cdot d\mathbf{S} = \iint_S F \cdot \underline{n} dS$. [6 marks]

B6. Use Stokes' Theorem to evaluate $\iint_S (\nabla \times F) \cdot \underline{n} dS$, where $F = -y\underline{i} + xz\underline{j} + x^2yz^3\underline{k}$, S is the part of $z = 6 - x^2 - y^2$ which is above the plane $z = 2$ and $\underline{n}$ is the outer normal vector to S . [10 marks]

This page to be used for continuation of your answers, if you need it. Be sure to indicate on any given question if your solution extends here.